

## **Cellular Expert**

### **8. RL Prediction**

## 1. Objective

This tutorial will show you how to add new MW links, manage and do predictions.

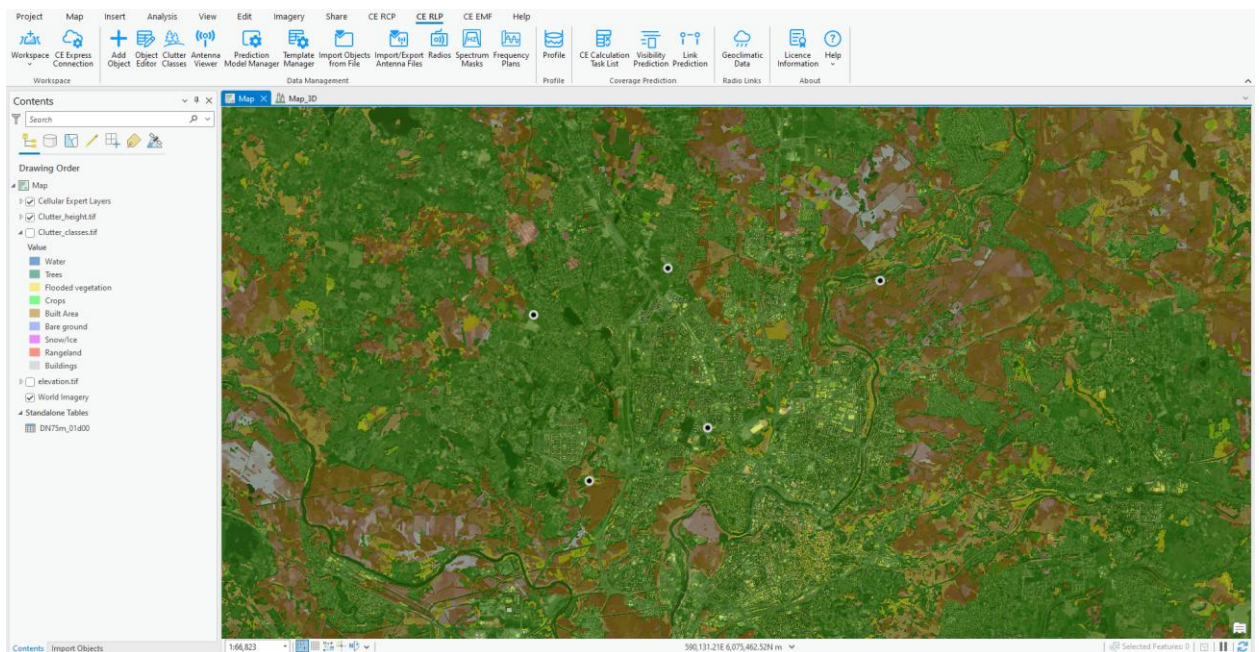
At the end of the exercise you will be able to:

- Create and import MW equipment.
- Create MW links in the project.
- Do MW Predictions.

## 2. Initial data

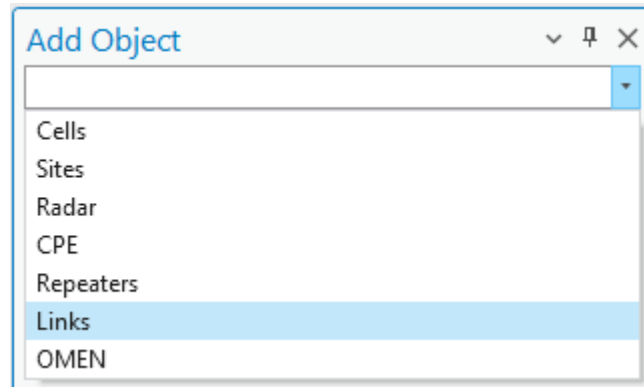
Prepared project with:

- Network objects.
- Geodata.
- Equipment and models.



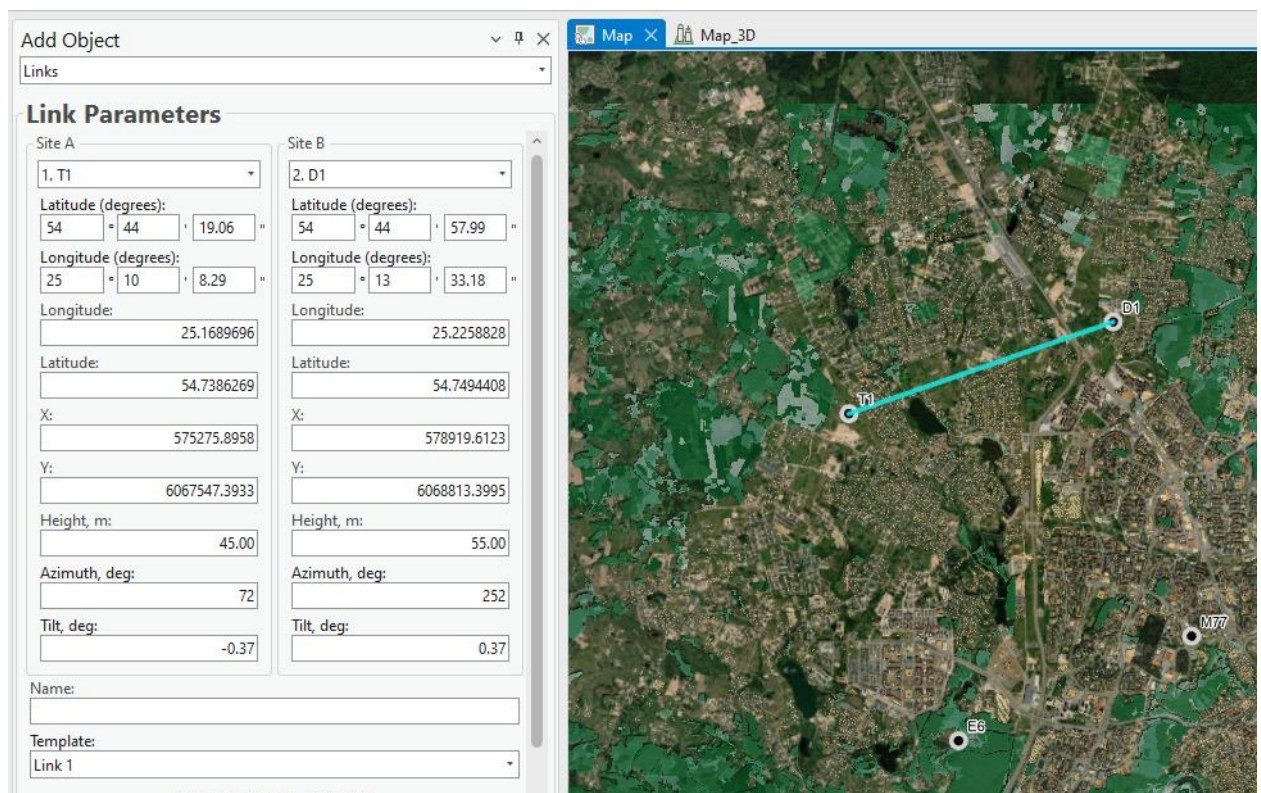
## 3. Add Links

The link objects can be imported or created manually using Add functionality. Open Add Object tool and select Link option from drop-down menu list.



Click once on T1 site and second time to D1 site.

It will fill general parameters automatically.

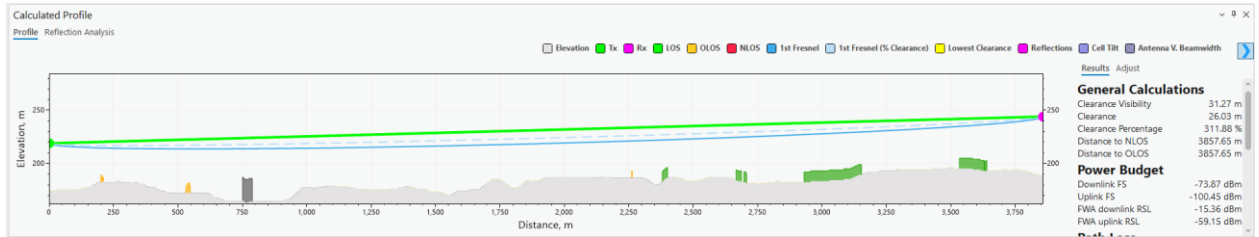


Height, azimuth and other parameters are calculated automatically, leave them as it is.

Press Show Profile button to preview topographical data between Site A and Site B, if there is LOS, what Fresnel zone percentage value, etc.



Profile will be generated.



Close Profile windows.

Define parameters are defined below:

- Name: MW001
- Radio Model: 2. Aurora 2400

Site A:  Site B:

Radio Model:

- Frequency Plan: 10GHz-CEPT12-05-3\_5MHz-350MHz

Frequency Plans:

Default Frequency Plan

2\_5-2\_7GHz-ITU-RF\_283-2GHz-119MHz

4GHz-ITU-RF\_635Ann1par\_3-40MHz-320MHz

7\_4-7\_7GHz-ITU-RF\_385-7GHzmainbody-7MHz-161MHz

- Carriers: 1 and 3 (1' and 3' will be automatically assigned).

Lower Upper

Selected carriers: 1, 1', 3, 3'

| Selected                            | Frequency, M | Power, dBm | Polarization |            |
|-------------------------------------|--------------|------------|--------------|------------|
| <input checked="" type="checkbox"/> | 1            | 10152.25   | 18           | Horizontal |
| <input type="checkbox"/>            | 2            | 10155.75   | 18           | Horizontal |
| <input checked="" type="checkbox"/> | 3            | 10159.25   | 18           | Horizontal |

- Go to Antenna section and change antenna to Antenna 10MHz

Frequency Plan Antenna

Equipment Pred. Models Performance

Site A

▼ Antenna: Antenna 10GHz View Antenna

| Objec | Manur         | Mode        | Frequ | Gain, |
|-------|---------------|-------------|-------|-------|
| 16    | ANDREW CC     | UHX8-127    | 12700 | 46    |
| 532   | Tutorial Radi | Antenna 10C | 10000 | 38    |

Misc Loss, dB

0.00

Site B

▼ Antenna: Antenna 10GHz View Antenna

| Objec | Manur         | Mode        | Frequ | Gain, |
|-------|---------------|-------------|-------|-------|
| 16    | ANDREW CC     | UHX8-127    | 12700 | 46    |
| 532   | Tutorial Radi | Antenna 10C | 10000 | 38    |

Misc Loss, dB

0.00

Press Save Changes button.

Do not close the dialog, create new links and define parameters:

- Between D1 and E2
  - Name: MW002
  - SiteA: Lower

Site A: Lower Site B: Upper

- Radio Model: 2. Aurora 2400

Site A: Upper Site B: Lower

Radio Model:

2. Aurora 2400

- Frequency Plan: 10GHz-CEPT12-05-3\_5MHz-350MHz

Frequency Plans:

10GHz-CEPT12-05-3\_5MHz-350MHz

Default Frequency Plan

2\_5-2\_7GHz-ITU-RF\_283-2GHz-119MHz

4GHz-ITU-RF\_635Ann1par\_3-40MHz-320MHz

7\_4-7\_7GHz-ITU-RF\_385-7GHzmainbody-7MHz-161MHz

10GHz-CEPT12-05-3\_5MHz-350MHz

- Carriers: 2 and 4

Lower Upper

Selected carriers: 2, 2', 4, 4'

| Selected                              | Frequency, MHz | Power, dBm | Polarization |
|---------------------------------------|----------------|------------|--------------|
| <input checked="" type="checkbox"/> 2 | 10155.75       | 18         | Horizontal   |
| <input type="checkbox"/> 3            | 10159.25       | 18         | Horizontal   |
| <input checked="" type="checkbox"/> 4 | 10162.75       | 18         | Horizontal   |

- Antenna for Site A and Site B: Antenna 10MHz

Press Save Changes

- Between E6 and M77
  - Name: MW003
  - SiteA: Lower

Site A: Lower Site B: Upper

- Radio Model: 2. Aurora 2400

Site A: Upper Site B: Lower

Radio Model:

2. Aurora 2400

- Frequency Plan: 10GHz-CEPT12-05-3\_5MHz-350MHz

Frequency Plans:

10GHz-CEPT12-05-3\_5MHz-350MHz

Default Frequency Plan

2\_5-2\_7GHz-ITU-RF\_283-2GHz-119MHz

4GHz-ITU-RF\_635Ann1par\_3-40MHz-320MHz

7\_4-7\_7GHz-ITU-RF\_385-7GHzmainbody-7MHz-161MHz

10GHz-CEPT12-05-3\_5MHz-350MHz

- Carriers: 1 and 4



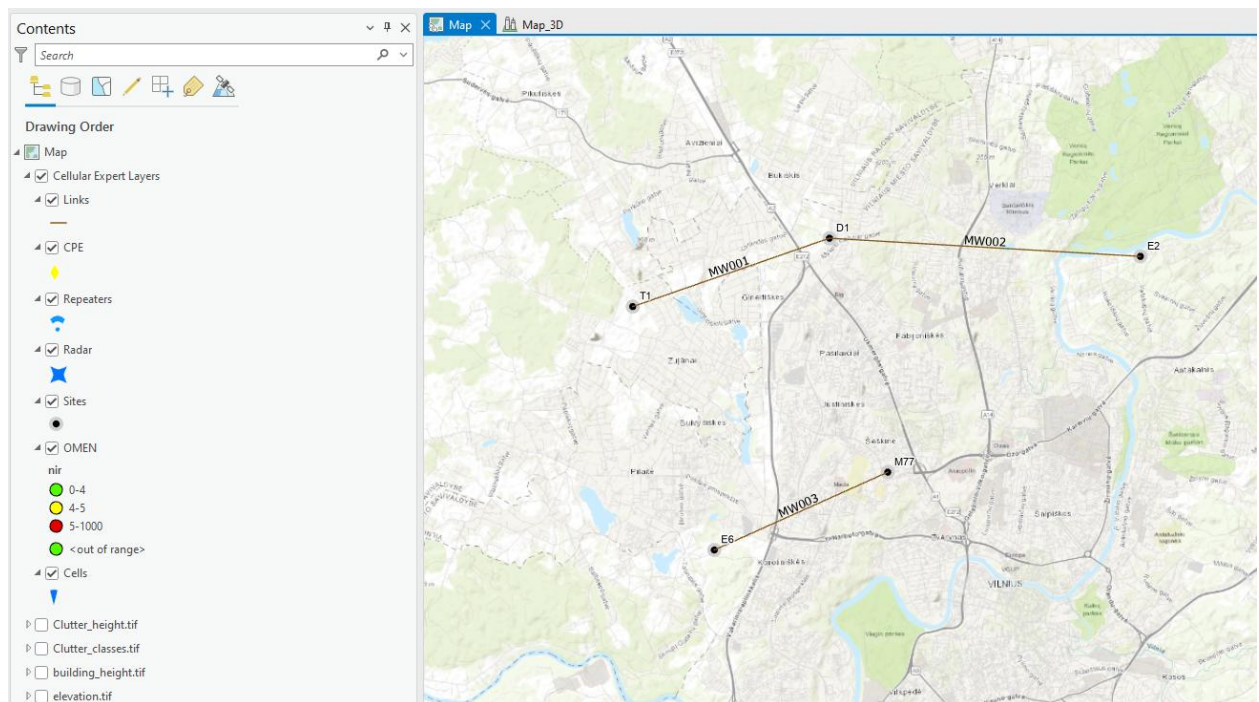
Lower Upper

Selected carriers: 1, 1', 4, 4'

| Selected                              | Frequency, M | Power, dBm | Polarization |
|---------------------------------------|--------------|------------|--------------|
| <input checked="" type="checkbox"/> 1 | 10152.25     | 18         | Horizontal ▾ |
| <input type="checkbox"/> 2            | 10155.75     | 18         | Horizontal ▾ |
| <input type="checkbox"/> 3            | 10159.25     | 18         | Horizontal ▾ |

- Antenna for Site A and Site B: Antenna 10MHz

Press Save Changes



## 4. RL Predictions

The predictions are working between selected links and provides all necessary information about power budget, interference or availability.

Select all links on the map and open Link Prediction tool. Before launching the predictions, enable Calculate Interference option.

### Link Prediction

Calculation Tools Export Options

**Calculation settings**

Calculation name

Link prediction

Link template

Link 10 GHz Duplex

☒ Calculate Interference

☐ Use Interference Radius

Interference Radius, m

1000

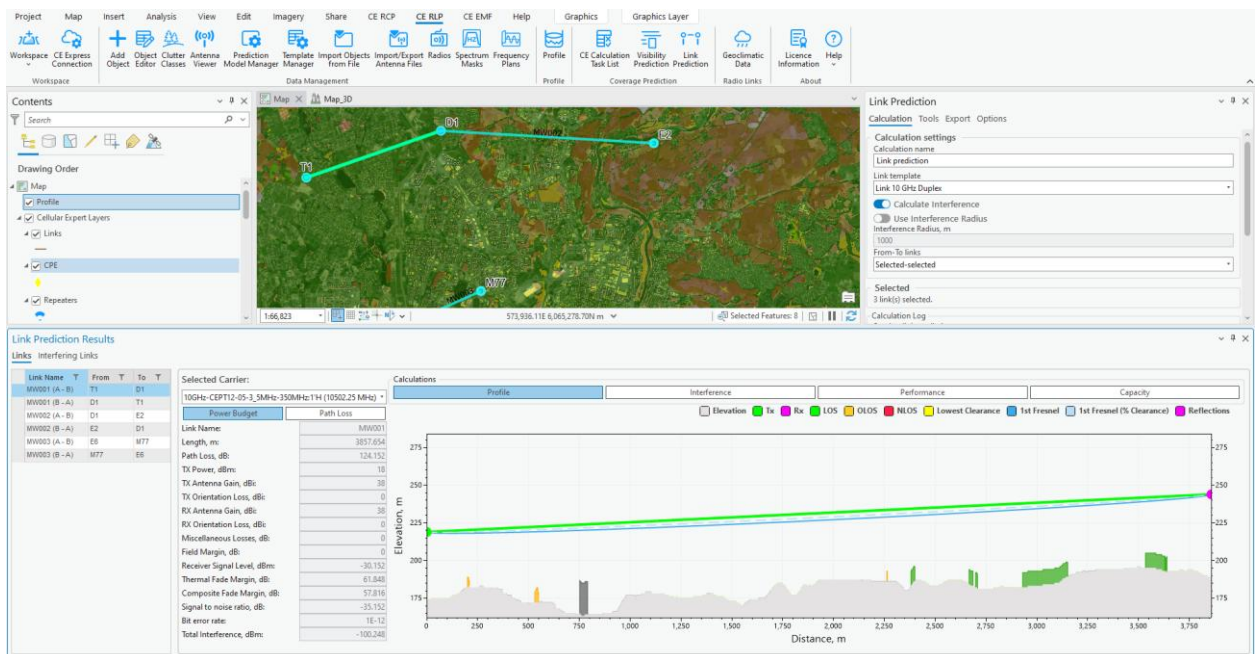
From-To links

Selected-selected

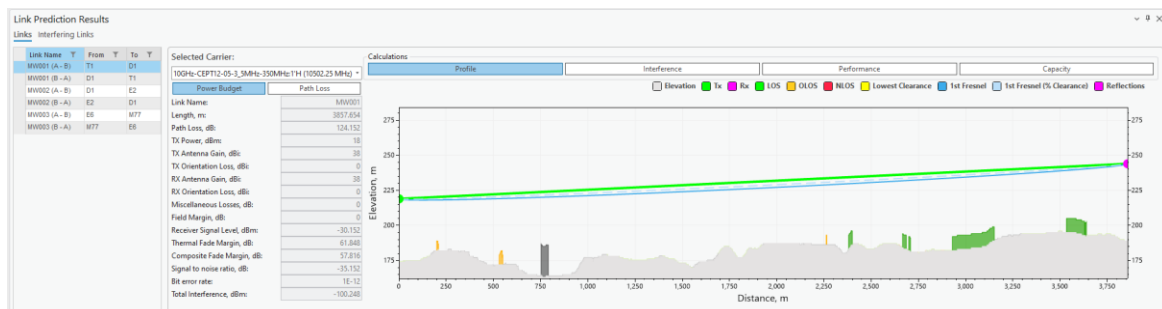
**Selected**

3 link(s) selected.

Press Run button and wait will prediction are completed and results loaded into your project.

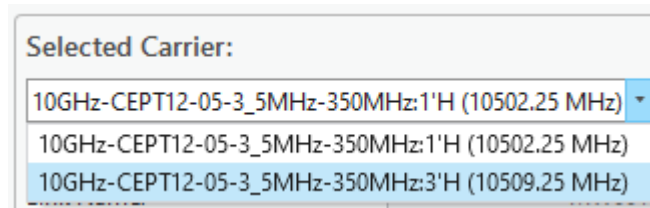


The primary results are Power Budget and Profile between Tx and Rx. On the left, select another link and results will be updated accordingly.





Selected Link can have several carriers, to preview the results expand Carrier option.



Click on Interference tab to display interfering links and interfered links information.

| Link Prediction Results |     |      |    |   |  |  |  |  |  |
|-------------------------|-----|------|----|---|--|--|--|--|--|
| Links Interfering Links |     |      |    |   |  |  |  |  |  |
| Link Name               | T   | From | To | T |  |  |  |  |  |
| MW001 (A-B)             | T1  | D1   |    |   |  |  |  |  |  |
| MW001 (B-A)             | D1  | T1   |    |   |  |  |  |  |  |
| MW002 (A-B)             | D1  | E2   |    |   |  |  |  |  |  |
| MW002 (B-A)             | E2  | D1   |    |   |  |  |  |  |  |
| MW003 (A-B)             | E6  | M77  |    |   |  |  |  |  |  |
| MW003 (B-A)             | M77 | E6   |    |   |  |  |  |  |  |

| Calculations                                                       |          |  |  |  |           |           |                            |           |          |
|--------------------------------------------------------------------|----------|--|--|--|-----------|-----------|----------------------------|-----------|----------|
| Selected Carrier: 10GHz-CEPT12-05-3_5MHz-350MHz:1'H (10502.25 MHz) |          |  |  |  |           |           |                            |           |          |
| Power Budget                                                       |          |  |  |  | Path Loss |           |                            |           |          |
| Link Name:                                                         | MW001    |  |  |  | Link Name | Site Name | Channel                    | Length, m | Loss, dB |
| Length, m:                                                         | 3857.654 |  |  |  | MW002     | E2        | 10GHz-CEPT12-05-3_5MHz-350 | 5771.847  | 127.655  |
| Path Loss, dB:                                                     | 124.152  |  |  |  | MW002     | E2        | 10GHz-CEPT12-05-3_5MHz-350 | 5771.847  | 127.655  |
| TX Power, dBm:                                                     | 18       |  |  |  | MW003     | E6        | 10GHz-CEPT12-05-3_5MHz-350 | 6153.163  | 127.913  |
| TX Antenna Gain, dBi:                                              | 38       |  |  |  | MW003     | E6        | 10GHz-CEPT12-05-3_5MHz-350 | 6153.163  | 127.922  |
| TX Orientation Loss, dBi:                                          | 0        |  |  |  | MW003     | M77       | 10GHz-CEPT12-05-3_5MHz-350 | 4464.839  | 125.422  |
| RX Antenna Gain, dBi:                                              | 38       |  |  |  | MW003     | M77       | 10GHz-CEPT12-05-3_5MHz-350 | 4464.839  | 125.43   |
| RX Orientation Loss, dBi:                                          | 0        |  |  |  |           |           |                            |           |          |
| Miscellaneous Losses, dB:                                          | 0        |  |  |  |           |           |                            |           |          |
| Field Margin, dB:                                                  | 0        |  |  |  |           |           |                            |           |          |
| Receiver Signal Level, dBm:                                        | -30.152  |  |  |  |           |           |                            |           |          |
| Thermal Fade Margin, dB:                                           | 61.848   |  |  |  |           |           |                            |           |          |
| Composite Fade Margin, dB:                                         | 57.816   |  |  |  |           |           |                            |           |          |
| Signal to noise ratio, dB:                                         | -35.152  |  |  |  |           |           |                            |           |          |
| Bit error rate:                                                    | 1E-12    |  |  |  |           |           |                            |           |          |
| Total Interference, dBm:                                           | -100.248 |  |  |  |           |           |                            |           |          |

| Interference from (Interfering links): |           |                            |           |          |                   |         |            |  |  |
|----------------------------------------|-----------|----------------------------|-----------|----------|-------------------|---------|------------|--|--|
| Link Name                              | Site Name | Channel                    | Length, m | Loss, dB | Interference, dBm | SIR, dB | FML, dB    |  |  |
| MW001                                  | E2        | 10GHz-CEPT12-05-3_5MHz-350 | 3451.768  | 162.368  | -228.099          | 186.738 | 0          |  |  |
| MW001                                  | E2        | 10GHz-CEPT12-05-3_5MHz-350 | 3451.768  | 162.368  | -228.099          | 186.732 | 0          |  |  |
| MW001                                  | D1        | 10GHz-CEPT12-05-3_5MHz-350 | 3857.654  | 124.152  | -96.749           | 63.69   | 2.866E-010 |  |  |
| MW001                                  | D1        | 10GHz-CEPT12-05-3_5MHz-350 | 3857.654  | 124.152  | -141.595          | 197.834 | 9.643E-010 |  |  |
| MW001                                  | M77       | 10GHz-CEPT12-05-3_5MHz-350 | 5832.582  | 127.44   | -213.948          | 184.884 | 0          |  |  |
| MW001                                  | M77       | 10GHz-CEPT12-05-3_5MHz-350 | 5832.582  | 127.44   | -213.948          | 184.875 | 0          |  |  |
| MW001                                  | E6        | 10GHz-CEPT12-05-3_5MHz-350 | 4752.913  | 125.965  | -144.721          | 115.362 | 4.822E-010 |  |  |

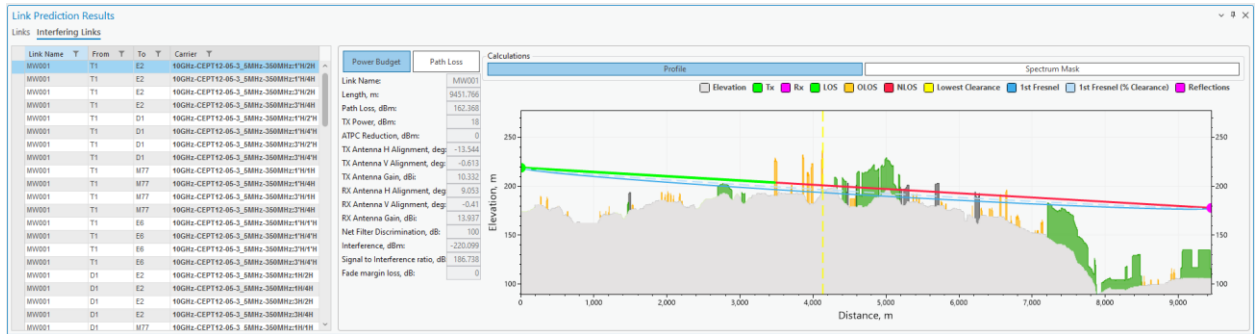
  

| Interference to (Interfered links): |           |                            |           |          |                   |         |            |  |  |
|-------------------------------------|-----------|----------------------------|-----------|----------|-------------------|---------|------------|--|--|
| Link Name                           | Site Name | Channel                    | Length, m | Loss, dB | Interference, dBm | SIR, dB | FML, dB    |  |  |
| MW001                               | E2        | 10GHz-CEPT12-05-3_5MHz-350 | 3451.768  | 162.368  | -228.099          | 186.738 | 0          |  |  |
| MW001                               | E2        | 10GHz-CEPT12-05-3_5MHz-350 | 3451.768  | 162.368  | -228.099          | 186.732 | 0          |  |  |
| MW001                               | D1        | 10GHz-CEPT12-05-3_5MHz-350 | 3857.654  | 124.152  | -96.749           | 63.69   | 2.866E-010 |  |  |
| MW001                               | D1        | 10GHz-CEPT12-05-3_5MHz-350 | 3857.654  | 124.152  | -141.595          | 197.834 | 9.643E-010 |  |  |
| MW001                               | M77       | 10GHz-CEPT12-05-3_5MHz-350 | 5832.582  | 127.44   | -213.948          | 184.884 | 0          |  |  |
| MW001                               | M77       | 10GHz-CEPT12-05-3_5MHz-350 | 5832.582  | 127.44   | -213.948          | 184.875 | 0          |  |  |
| MW001                               | E6        | 10GHz-CEPT12-05-3_5MHz-350 | 4752.913  | 125.965  | -144.721          | 115.362 | 4.822E-010 |  |  |

Total Interference is shown in Power Budget section.

| Selected Carrier:                                |           |
|--------------------------------------------------|-----------|
| 10GHz-CEPT12-05-3_5MHz-350MHz:1'H (10502.25 MHz) |           |
| Power Budget                                     | Path Loss |
| Link Name:                                       | MW001     |
| Length, m:                                       | 3857.654  |
| Path Loss, dB:                                   | 124.152   |
| TX Power, dBm:                                   | 18        |
| TX Antenna Gain, dBi:                            | 38        |
| TX Orientation Loss, dBi:                        | 0         |
| RX Antenna Gain, dBi:                            | 38        |
| RX Orientation Loss, dBi:                        | 0         |
| Miscellaneous Losses, dB:                        | 0         |
| Field Margin, dB:                                | 0         |
| Receiver Signal Level, dBm:                      | -30.152   |
| Thermal Fade Margin, dB:                         | 61.848    |
| Composite Fade Margin, dB:                       | 57.816    |
| Signal to noise ratio, dB:                       | -35.152   |
| Bit error rate:                                  | 1E-12     |
| Total Interference, dBm:                         | -100.248  |

Interference link predictions can be previewed too. Click on Interfering Link in top left corner and it will display selected interfering link Power Budget, Path Loss and draw a line on the map.



Go back to Links section, select MW001 from T1 to D1, Carrier 1'H.

**Link Prediction Results**

**Links** Interfering Links

| Link Name     | From | To  |
|---------------|------|-----|
| MW001 (A - B) | T1   | D1  |
| MW001 (B - A) | D1   | T1  |
| MW002 (A - B) | D1   | E2  |
| MW002 (B - A) | E2   | D1  |
| MW003 (A - B) | E6   | M77 |
| MW003 (B - A) | M77  | E6  |

**Selected Carrier:**  
10GHz-CEPT12-05-3\_5MHz-350MHz:1'H (10502.25 MHz)

**Power Budget** **Path Loss**

Link Name: MW001  
Length, m: 3857.654  
Path Loss, dB: 124.152  
TX Power, dBm: 18

Click on Interference tab, and analyze Inteferece From table. Link MW002, from Site: E2 with Carrier 2'H interferences:

- Interference, dBm: -100.248 dBm
- FML, dB: 1.29E-10

Interference from (interfering links):

| Link Name | Site Name | Channel                           | Length, m | Loss, dB | Interference, dBm | SIR, dB | FML, dB    |
|-----------|-----------|-----------------------------------|-----------|----------|-------------------|---------|------------|
| MW002     | E2        | 10GHz-CEPT12-05-3_5MHz-350MHz:2'H | 5771.847  | 127.655  | -100.248          | 70.096  | 1.297E-010 |

Close Link Prediction results, select MW002 link on the map and open Object Editor.

Double click on the link.

**Object Editor**

Move Objects Duplicate Objects

Links  
MW002 (Object ID: 2)

**Link Parameters**

Site A: 2. D1  
Latitude (degrees): 54.44 57.99

Site B: 3. E2  
Latitude (degrees): 54.44 43.73

Find Frequency Plan table, select Upper and change carrier 2' polarization to Vertical.

Lower Upper

Selected carriers: 2, 2', 4, 4'

| Selected                               | Frequency, MHz | Power, dBm | Polarization |
|----------------------------------------|----------------|------------|--------------|
| <input type="checkbox"/> 1'            | 10502.25       | 18         | Horizontal   |
| <input checked="" type="checkbox"/> 2' | 10505.75       | 18         | Vertical     |
| <input type="checkbox"/> 3'            | 10509.25       | 18         | Horizontal   |

Save Changes.

Select all links on the map, open Link Prediction tool again and run predictions.

After predictions are done, open MW001 (A-B) interference predictions and preview how Interference From is changed for MW002 link, Site E2.

| Interference from (interfering links): |           |                                   |           |          |                   |         |            |
|----------------------------------------|-----------|-----------------------------------|-----------|----------|-------------------|---------|------------|
| Link Name                              | Site Name | Channel                           | Length, m | Loss, dB | Interference, dBm | SIR, dB | FML, dB    |
| MW002                                  | E2        | 10GHz-CEPT12-05-3_5MHz-350MHz 2'V | 5771.847  | 127.655  | -130.248          | 100.096 | 1.302E-013 |

Close Link Prediction Results dialog.

#### 4.1 Geoclimatic data

Geoclimatic data, comprising factors such as rain rate, temperature, gaseous absorption, and multipath fading, significantly shapes the planning and efficacy of microwave links. Rain rate introduces the challenge of rain fade, where increased rainfall leads to greater signal attenuation. To address this, microwave link planners employ rain attenuation models, adjusting transmitter power or incorporating diversity techniques to maintain link reliability. Temperature variations impact atmospheric conditions, influencing signal propagation. Planners consider these effects to optimize link performance and account for temperature-related changes in their designs. Gaseous absorption, particularly by atmospheric water vapor and oxygen, is factored into link planning by selecting frequency bands less susceptible to absorption, particularly for long-distance links. Lastly, the impact of multipath fading is mitigated through strategies like antenna diversity and adaptive modulation, as well as terrain-aware antenna placement. By navigating these geoclimatic challenges, microwave link planners ensure robust and effective communication links in diverse environmental conditions.

The tool is located in CE RLP tab, Radio Links section.

Run Link Predictions again. After successful calculations, leave MW001 (A-B) link selected, and click on Performance tab.

| Calculations                                                                                                                                                                                                                                             |                     |                      |                      |                       |                             |                              |                 |                  |                   |                    |                       |                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------|----------------------|-----------------------|-----------------------------|------------------------------|-----------------|------------------|-------------------|--------------------|-----------------------|------------------|
| Profile                                                                                                                                                                                                                                                  |                     |                      | Interference         |                       |                             | Performance                  |                 |                  | Capacity          |                    |                       |                  |
| <div><div><div><input checked="" type="radio"/> Propagation reliability</div><div><input type="radio"/> Total reliability</div></div><div><div>Diversity improvement</div><div>1.00</div><div>Protection improvement</div><div>4166.33</div></div></div> |                     |                      |                      |                       |                             |                              |                 |                  |                   |                    |                       |                  |
|                                                                                                                                                                                                                                                          | Total Outage Annual | Total Outage W.Month | Multipath ITU Annual | Multipath ITU W.Month | Multipath V.-Barnett Annual | Multipath V.-Barnett W.Month | Rain ITU Annual | Rain ITU W.Month | Rain Crane Annual | Rain Crane W.Month | Multipath XPD W.Month | Rain XPD W.Month |
| Unavailability, %                                                                                                                                                                                                                                        | 0.00018             | 0.00584              | 0.00000              | 0.00000               | 0.00000                     | 0.00000                      | 0.00018         | 0.00073          | 0.00000           | 0.00000            | 0.00511               | 0.00000          |
| Availability, %                                                                                                                                                                                                                                          | 99.99982            | 99.99416             | 100.00000            | 100.00000             | 100.00000                   | 100.00000                    | 99.99982        | 99.99927         | 100.00000         | 100.00000          | 99.99489              | 100.00000        |
| Objective, %                                                                                                                                                                                                                                             | 99.99800            |                      | 99.99900             |                       | 99.99900                    |                              | 99.99900        |                  | 99.99900          |                    |                       |                  |
| Unavailability time                                                                                                                                                                                                                                      | 0:00:57             | 0:30:42              | 0:00:00              | 0:00:00               | 0:00:00                     | 0:00:00                      | 0:00:57         | 0:03:49          | 0:00:00           | 0:00:00            | 0:26:53               | 0:00:00          |
| Availability time                                                                                                                                                                                                                                        | 8765:59:03          | 8765:29:18           | 8766:00:00           | 8766:00:00            | 8766:00:00                  | 8766:00:00                   | 8765:59:03      | 8765:56:11       | 8766:00:00        | 8766:00:00         | 8765:33:07            | 8766:00:00       |
| Objective time                                                                                                                                                                                                                                           | 0:00:00             |                      | 8765:54:44           |                       | 8765:54:44                  |                              | 8765:54:44      |                  | 8765:54:44        |                    |                       |                  |

Review Multipath ITU and Rain ITU percentage values.

| Multipath ITU Annual | Multipath ITU W.Month | Rain ITU Annual | Rain ITU W.Month |
|----------------------|-----------------------|-----------------|------------------|
| 0.00000              | 0.00000               | 0.00018         | 0.00073          |
| 100.00000            | 100.00000             | 99.99982        | 99.99927         |
| 99.99900             |                       | 99.99900        |                  |
| 0:00:00              | 0:00:00               | 0:00:57         | 0:03:49          |
| 8766:00:00           | 8766:00:00            | 8765:59:03      | 8765:56:11       |
| 8765:54:44           |                       | 8765:54:44      |                  |

Close results, and open Geoclimatic Data tool.

Geoclimatic Data

Gaseous Absorption

Temperature

Multipath Fading

Rain Fading

Statistics

Dry air pressure, hPa

Water vapour density, g/m3

☒ Use geoclimatic data

Water vapour density data

Water vapour density field

Absorption Spectrum

Click on Multipath Fading tab, and define:

- ITU-R P.530 version: 17

Geoclimatic Data

Gaseous Absorption Temperature

Multipath Fading Rain Fading Statistics

☒ ITU method

ITU-R P.530 version: 17

Click on Rain Fading tab, and define:

Geoclimatic Data

Gaseous Absorption Temperature

Multipath Fading Rain Fading Statistics

Rain fall data

☐ Rain zone

Category: ITU Zone: E

ITU - E

☒ Geoclimatic data

ESA rain rate data: ESARAIN ESA rain rate field: MT

Fading method

☒ ITU method

ITU-R P.530 version: 18

Rain rate exceeded for 0.01% of the time (mm/h)

☐ Use default 22 ☒ Use custom 50

☒ Crane

End-to-end reliability method

Multi-hop: Accumulation Two-way: Accumulation

Press Save Changes. Close Geoclimatic Data tab, and run Link Predictions again. Review Performance tab calculations for MW001 (A-B) link. Compare them with previous results.